5.9 Applying Ratios to Solve Problems Involving Percentages

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<tr><td>Benchmark</td><td colspan="2">MA.6.AR.3.4 Apply ratio relationships to solve mathematical and real-world problems involving percentages using the relationship between two quantities.</td><td>Learning Target</td><td>• I can apply ratio relationships to solve problems involving percentages.</td></tr>
<tr><td rowspan="2">Benchmark Coherence</td><td colspan="2">Building On</td><td colspan="2">Working Toward</td></tr>
<tr><td colspan="2">N/A</td><td colspan="2">MA.7.AR.3.1 Apply previous understanding of percentages and ratios to solve multi-step real-world percent problems.</td></tr>
<tr><td>Essential Questions</td><td colspan="4">• How do you apply ratio relationships to solve problems involving percentages?
• In a real-world context, how do you identify the value for the part?
• In a real-world context, how do you identify the value for the whole?</td></tr>
<tr><td>Lesson Narrative</td><td colspan="4">In this lesson, students extend their understanding of ratios and percentages to apply them to real-world contexts. Students begin by using double number lines, tape diagrams, and tables to find equivalent ratios and their corresponding percentages. Students continue to build comprehension through using the more abstract method of representation, tables, to find missing values and percentages based on a context and ratio. The lesson culminates in students demonstrating their ability to use equivalent ratio tables to calculate percentages in real world contexts.</td></tr>
<tr><td rowspan="3">Component Summary</td><td>5.9.1 Warm-Up (~5 min.)</td><td>Use a double number line to estimate percentages (SE p. 214)</td><td>5.9.3 Exploration (15-20 min.)</td><td>Explore using equivalent ratio tables to calculate percentages in real-world contexts (SE p. 216)</td></tr>
<tr><td>5.9.2 Guided Practice (5-10 min.)</td><td>Apply ratio relationships to solve real-world percentage problems (SE p. 215)</td><td>5.9.4 Your Turn! (10 min.)</td><td>Practice using equivalent ratio tables to calculate percentages in real-world contexts (SE p. 217)</td></tr>
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<u>Additional Terms</u>: N/A</td><td colspan="2">N/A</td><td>N/A</td></tr>
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Teacher Tips	Common Misconceptions	Things to Consider
	• Students may accidentally multiply on one side and divide on the other side of the equivalent ratio table, as they may do in solving equations. • Students may think that the equivalent ratio in tables must be a whole number (see "fractions of a person" question in 5.9.3 Exploration).	• Utilize data or informal observations from Lesson 8 to drive decision-making in instruction and practice for this lesson, as this lesson is primarily building on the knowledge on Lesson 8 in practice for Lesson 10. • Look at Lesson 11 when planning for this lesson to determine where your instruction and student practice can be spiraled.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder Consider pairing 5.9.3 Exploration with a Notice and Wonder routine. This will position students to make connections between ratios and percentages to navigate a variety of real-world contexts using different methods, particularly the methods and patterns that make the most sense to them as learners.	**MA.K12.MTR.2.1: Multiple Representations** Students apply ratios relationships to solve real-world problems involving percentages. Across the lesson components, the ratios are represented in seven equivalent ratio tables, two double number lines, and one tape diagram. Making connections among these representations deepens student understanding.
Differentiate Instruction	Consider using the double number line or equivalent ratio table to help students visualize how to write the ratio as a percentage in all components of the lesson. If one representation is provided in the lesson, consider also displaying the problem using the other representation.	

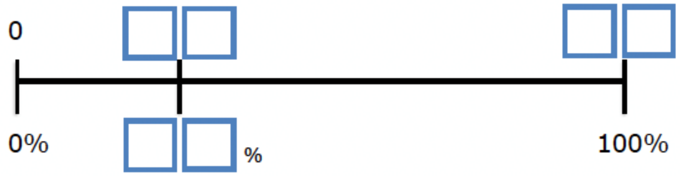
Lesson Notes:

5.9.1 Warm-Up: Percentages on the Number Line

Component Overview: *Students use a double number line to estimate percentages.*

5.9.1 Warm-Up: Percentages on the Number Line

1. Place a digit in each blue box to create an accurate number line. Use only digits 0 to 9, at most one time each. The number line is not drawn to scale.



Possible answers:
35%, 21, 60;
25% 09, 36;
34%, 17, 50

Students may wonder if the digits and resulting percentages should include decimals. Provide space for students to choose whether they think it should, as their responses will provide more entry points for learning.

5.9.2 Guided Practice: Applying Ratio Relationships to Solve Percentage Problems

Component Overview: *Students represent real-world ratio relationships with a table, double number line, and tape diagram. Have students complete each problem independently or with a partner, then choose representations to share with the class to provide students with an opportunity to check their work and discuss the different strategies.*

5.9.2 Guided Practice: Applying Ratio Relationships to Solve Percentage Problems

1. There is a 10% off sale on laptop computers. Ryan saved $35 off of the original laptop price during the sale. Use the ratio table to determine the original price of the laptop.

Dollar Amount	Percentage
35	10
350	100

× 10 × 10

2. Lily was running in her first 10K race in Islamorada, Florida. She asked her sister to cheer for her when she was 75% of the way finished with the race so that it would motivate her to finish strong. The race is approximately 6.2 miles long. Use the number line diagram to determine how many miles Lily will have run when she is 75% finished with the race.

4.65 *miles*

3. During the first part of his hike along the Myakka Hiking Trail in Florida State Park, Andre drank 1.5 liters of water. If this is 40% of the water he brought on the hike, use the tape diagrams to determine how much water he brought.

water (liters)	0.75	0.75	0.75	0.75	0.75
percentage	20	20	20	20	20

Andre brought 3.75 liters of water with him on the hike.

Provide additional scaffolding for students using Math Nation's On-Ramp to 6th Grade video on decimals and place values.

Activate prior knowledge and facilitate student connections from Lesson 8.

After students have completed each question, choose representations to share with the class and engage the class in a discussion. Consider asking questions to deepen conceptual understanding like:

- *How do each of the three representations accurately show each ratio?*
- *Explain which representation or model is the most efficient for you.*
- *How are the representations related?*
- *How can you use each of these representations to determine missing values or percentages?*

5.9.3 Exploration: Using Tables and Unit Rates With Percentages

Component Overview: *In this Exploration, students explore percentages using real-world scenarios. Students begin by working with a partner to describe work shown in a ratio table. Students then complete tables to find equivalent ratios and their corresponding percentages of the whole. Students then use the tables to complete statements to interpret the different scenarios. Allow students to work through this component on their own then debrief their work before moving on to 5.9.4 Your Turn!.*

5.9.3 Exploration: Using Tables and Unit Rates with Percentages

A restaurant has a sign by the front door that says, "Maximum occupancy: 75 people."

Damian completed the first two rows of a ratio table representing this situation. His work is shown.

$\times \frac{1}{75}$

Number of People	Percentage of Occupancy
75	100
1	$\frac{100}{75} = \frac{4}{3}$

$\times \frac{1}{75}$

If students have not mastered their multiplication and division facts, let them use a calculator to determine the ratios, rates, and percentages. The calculator will allow students to focus on mastering the learning goals of the lesson.

1. Work with your partner to complete the statements to describe Damian's work.

First, Damian interpreted the restaurant sign to mean that 100% is equal to 75 people.

Next, he multiplied both values in the ratio by $\frac{1}{75}$ to find the unit rate of the

- ○ percentage occupancy per 1 person
- ○ number of people per 1% occupancy

in the restaurant. He multiplied this factor by both values to find an equivalent ratio.

He then rewrote the result, $\frac{100}{75}$, as an equivalent fraction, $\frac{4}{3}$, by dividing both the numerator and denominator by 25.

2. With your partner, complete the table to find additional equivalent ratios. Use arrows and multiplication to show how you determine the values in each row.

	Number of People	Percentage of Occupancy	
	75	100	
$\times \frac{1}{75}$	1	$\frac{4}{3}$	$\times \frac{1}{75}$
$\times 9$	9	12	$\times 9$
$\times \frac{17}{3}$	51	68	$\times \frac{17}{3}$
$\times \frac{1}{68}$	$\frac{3}{4}$	1	$\times \frac{1}{68}$
$\times 40$	30	40	$\times 40$

3. Complete the statements to interpret the ratios.

When there are 9 people in the restaurant, it is 12 % full.

The restaurant is at 68 % capacity when 51 people are inside.

The restaurant is 40% full when 30 people are there.

4. Christian wondered if it is possible for the restaurant to ever be at exactly 1% capacity. Discuss his thinking with your partner. Then, write a brief summary of your discussion.

Answers will vary.

According to the ratio table, when the capacity of the restaurant is at 1%, there are ¾ of a person present. Answers should always be viable, and it is not possible to have ¾ of a person. This means that the restaurant will never have a capacity of 1%.

During this Exploration, circulate the room and observe students' informal discussions with their partners.

Listen for students who are using the formal vocabulary presented in previous lessons, such as unit rate and ratio, and connecting that to percentages. This illustrates conceptual understanding. Consider allowing these students to explain their work during the debrief at the end and highlighting the connections they have made.

Students may struggle to understand why the restaurant can never be at 1% capacity, because the idea of ¾ of a person may not make sense. Students may understand there can't be 1% capacity but may struggle to verbalize why. To help students verbalize their responses, ask probing questions like:

- *How can you determine how many people will be in the restaurant at 1% capacity?*
- *How can you use the table to help you find this?*
- *Would a fraction as the number of people be a reasonable answer here? Explain what you might do if the answer (number of people) results in a fraction.*

5.9.3 Exploration: Using Tables and Unit Rates With Percentages (continued)

Last Sunday, 1,575 people visited an amusement park.

- 56% of the visitors were adults
- 16% of the visitors were teenagers
- 28% of the visitors were children under the age of 12

5. Work with your partner to use the equivalent ratio table below to determine the number of adults, teenagers, and children who visited the park last Sunday. Show your work.
Answers will vary.

Number of Visitors	Percentage
1,575	100
15.75	1
882	56
252	16
441	28

$\times \frac{1}{100}$ $\times 56$ $\times 16$ $\times 28$

6. Complete the statements.

Last Sunday, 882 adults, 252 teenagers, and 441 children under the age of 12 visited the amusement park.

Notice and Wonder
Consider engaging the class in a "Notice and Wonder" routine before completing question 6 to solidify conceptual understanding. What patterns do you notice in this equivalent ratio table? What do you wonder about this equivalent ratio table?

5.9.4 Your Turn!

Component Overview: *Students work independently to practice using equivalent ratio tables to calculate percentages in real-world contexts.*

5.9.4 Your Turn!

1. At a hardware store, a tool set costs $80. Using a coupon, Denise is able to pay $12 less than the original price.

   a. Complete the table.

Dollar Amount	Percentage
$80	100
$12	15

$\times \frac{3}{20}$ $\times \frac{3}{20}$

   b. The coupon gives Denise 15% off of the original price.

2. A bathtub can hold 80 gallons of water. Isaiah fills it up to 65% capacity to take a bath.

   a. Complete the table to determine how many gallons of water Isaiah had in the bathtub.

Number of Gallons	Percentage of Capacity
80	100
52	65

$\times \frac{13}{20}$ $\times \frac{13}{20}$

   b. Isaiah filled the tub with _52_ gallons of water to be at 65% capacity.

The ratios in this table are more complex than they have been throughout the lesson, so students may struggle to determine what they need to multiply the top row by to find the second row. To help students work through this component, ask clarifying questions like:

- *Where can you fill in the information you already know on the table?*
- *How can you determine what to multiply 80 by to find 12? Will this be a whole number, or a fraction? How do you know?*
- *What percentage does 80 gallons represent? Why?*
- *What can you multiply 100 by to find 65?*

5.9.5 Wrap-Up: Ticket Out the Door

Component Overview: *Students demonstrate their ability to use equivalent ratio tables to calculate percentages in real-world contexts.*

5.9.5 Wrap-Up: Ticket Out the Door

1. A large school bus can hold a maximum of 72 children. This morning, Malachi's school bus only had 54 students on board. What percentage of Malachi's school bus is full?

Number of Students	Percentage of Capacity
72	100
54	75

$\times \frac{3}{4}$ $\times \frac{3}{4}$

Malachi's school bus was 75% full this morning.

Consider having students choose which representation they want to use to determine the percentage in substitution of completing the ratio table.